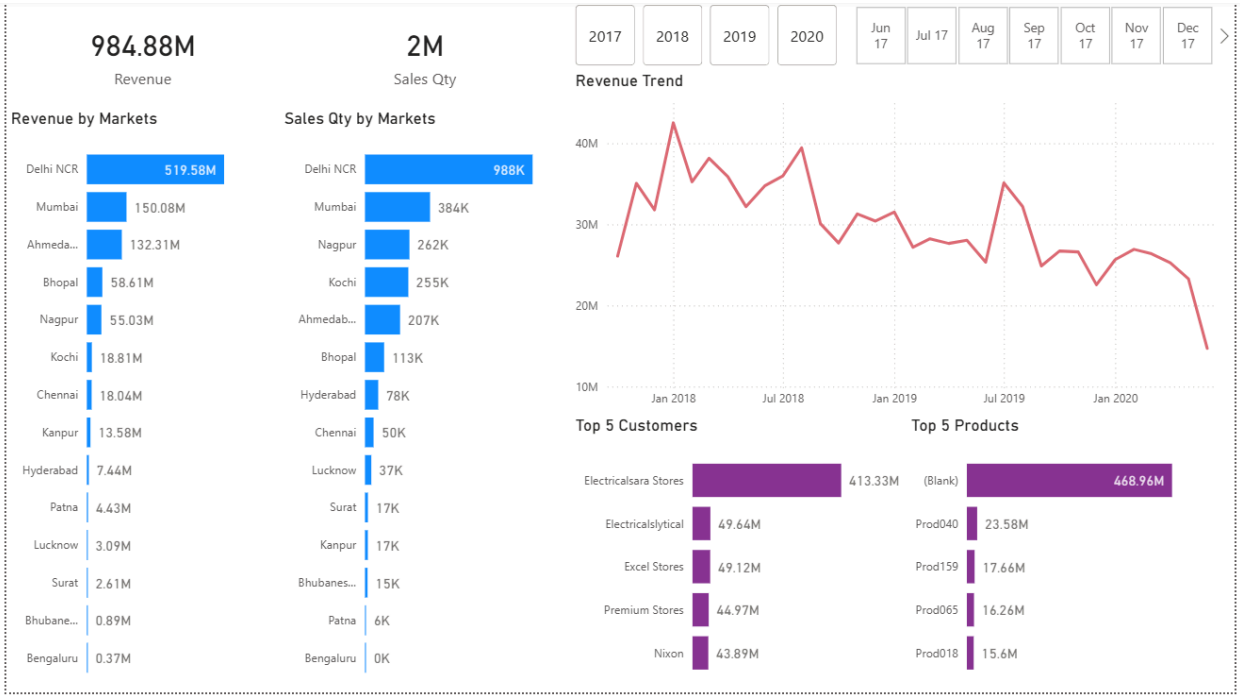
Building a dashboard for a hardware company

For this project, I am exploring dummy data in MySQL and Power BI to analyze sales and revenue trends for a hardware company.



Understanding the dataset

The analysis is built on five interrelated tables:

1. **Sales Transactions** is the core of the dataset, recording every individual sales event along with associated monetary, customer, product, and market information.
2. **Sales Products** serves as the product master table, providing classification details for each product sold.
3. **Sales Customers** contains the customer master data, enabling customer-level segmentation and analysis across transactions.
4. **Sales Markets** provides geographic context by grouping markets into broader zones, allowing for regional performance analysis.
5. **Sales Date** is a dedicated date dimension table that supports time-series analysis, enabling trends to be examined across days, months, and years.

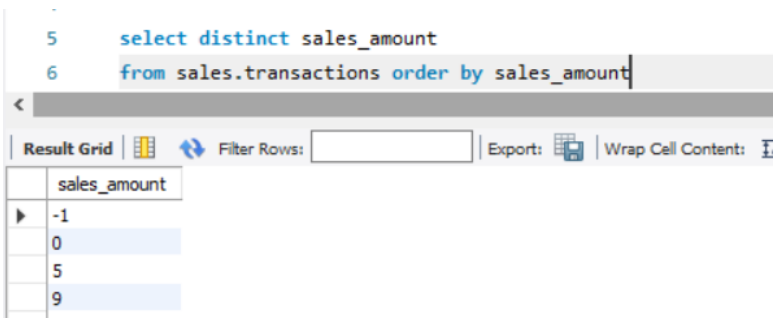
Exploratory questions

- Which products are performing well?
- Which areas are bringing in the most revenue?
- What is the revenue trend over time?

Data Cleaning

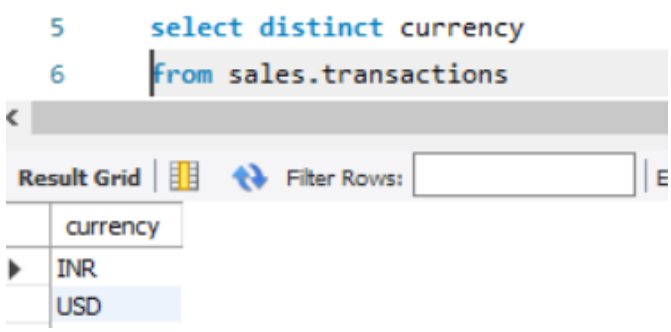
The following issues were noticed in the table:

1. sales_amount column in the Transactions table had values of 0, -1



   Solution: Filter out these values in Power Query.

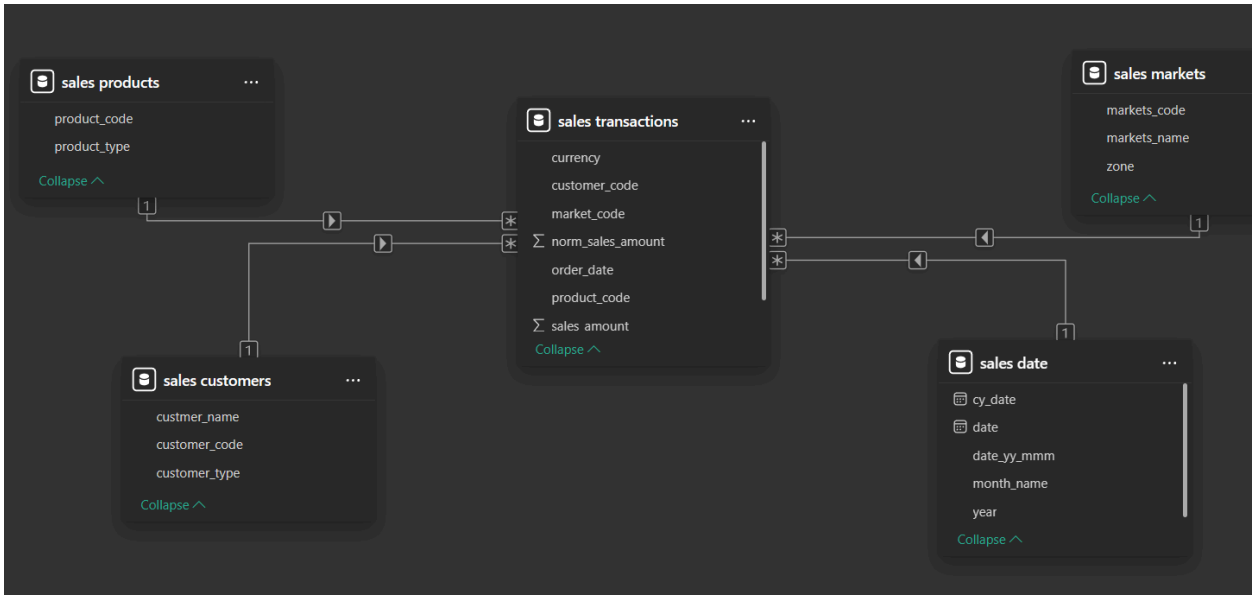
2. Two different currencies (INR and USD) were noticed in the table



   Solution: create a new column ‘normalized sales_amount’ with all amounts in INR only using Power Query in Power BI.

Data Modelling

The following relationships were made in Model View:



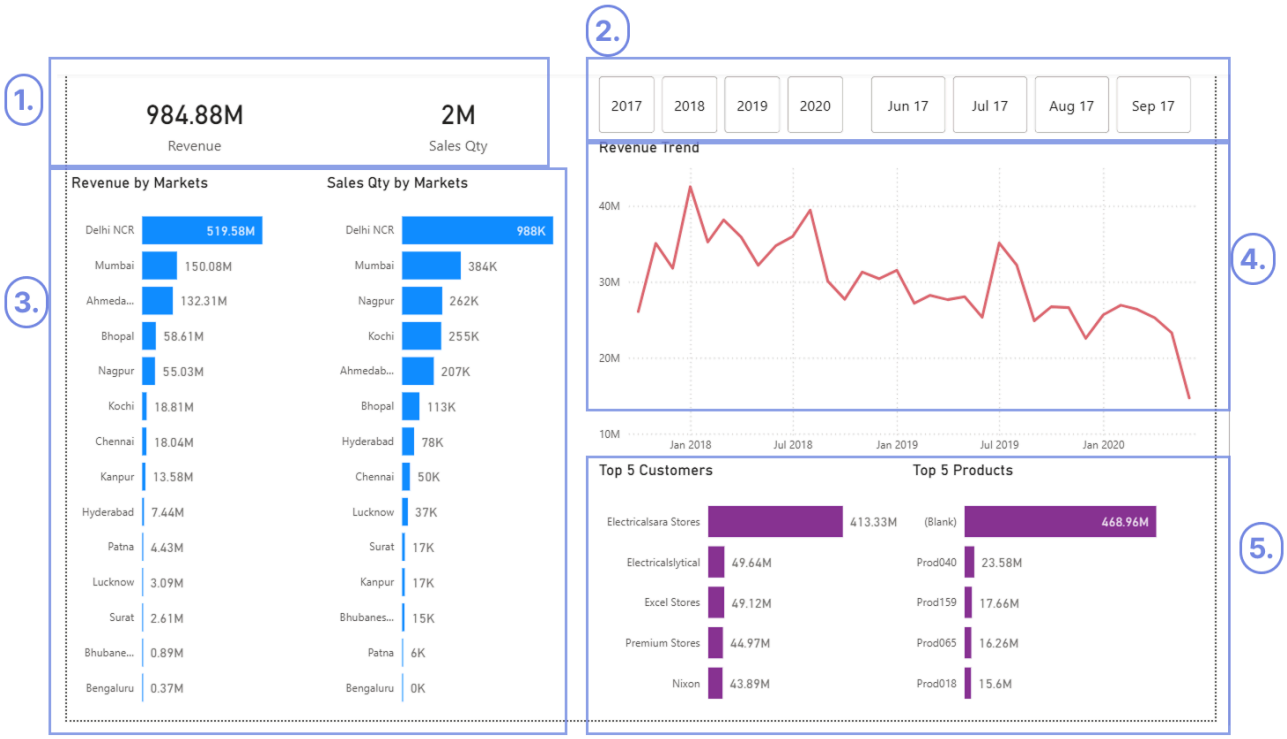
Data Visualization

Base Measures: Created the following aggregating measures -

```
Revenue = SUM('sales transactions'[norm_sales_amount])
```

```
Sales Qty = SUM('sales transactions'[sales_qty])
```

Charts and Visualizations:



1. Cards: To display the total revenue and total sales quantity.
2. Button Slicers (for year and month): For when someone wants to dive deeper into the trends for a particular year or month.
3. Bar charts:
    a. Market_name vs Revenue
    b. Market_name vs Sales quantity
4. Line chart: Cy_date (including year, quarter, month, day) vs Revenue
5. Bar charts:
    a. Top 5 customers bringing in most revenue
    b. Top 5 well performing products